

10

Correlation and Regression

10-1 Scatter Plots and Correlation

A **scatter plot** is a graph of the ordered pairs (x, y) of numbers consisting of the independent variable x and the dependent variable y .

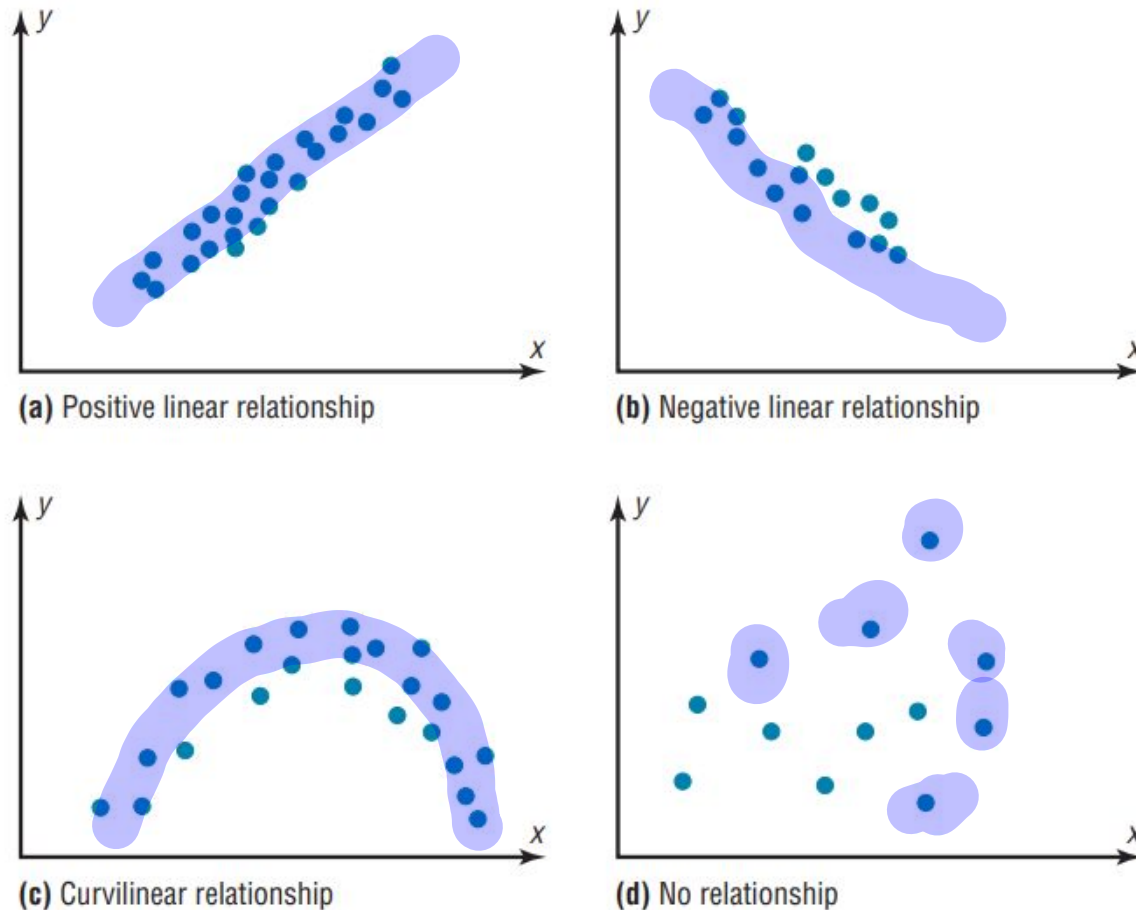


FIGURE 10-1
Types of Relationships

EXAMPLE 10-1 Car Rental Companies

Construct a scatter plot for the data shown for car rental companies in the United States for a recent year.

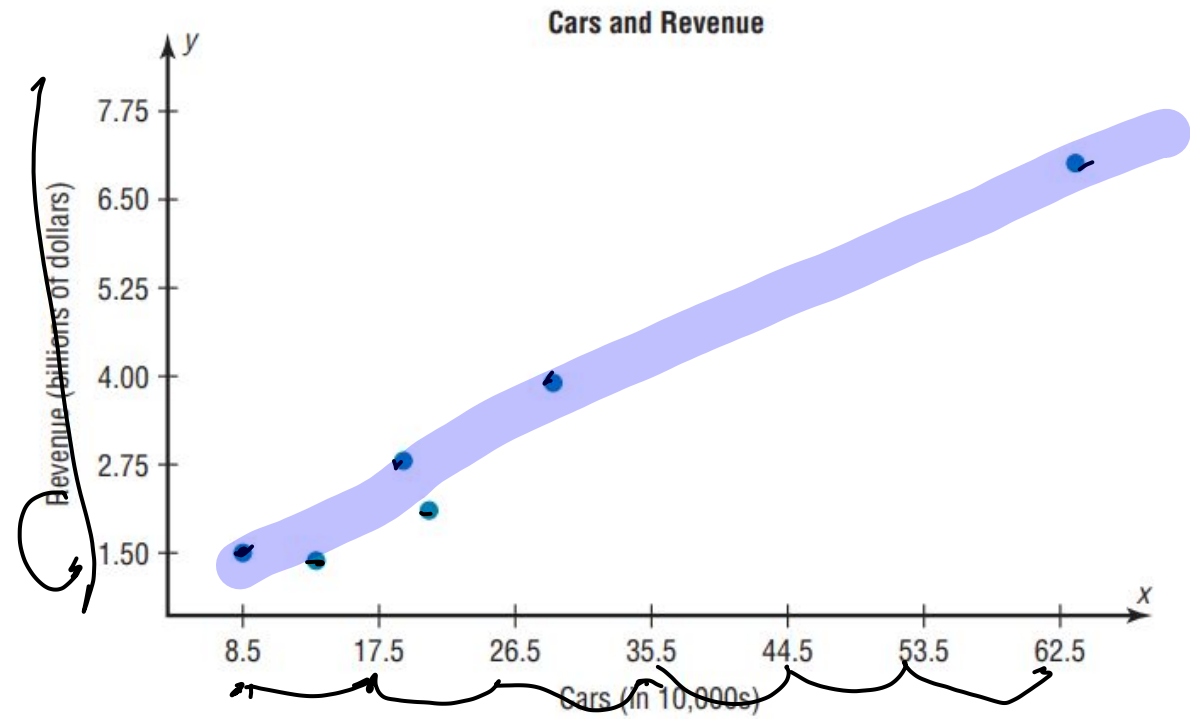
Company	Cars (in ten thousands)	Revenue (in billions)
A	63.0	\$7.0
B	29.0	3.9
C	20.8	2.1
D	19.1	2.8
E	13.4	1.4
F	8.5	1.5

Source: *Auto Rental News*.

SOLUTION

In this example, it looks as if a positive linear relationship exists between the number of cars that an agency owns and the total revenue that is made by the company.

FIGURE 10-2 Scatter Plot for Example 10-1



EXAMPLE 10-2 Absences and Final Grades

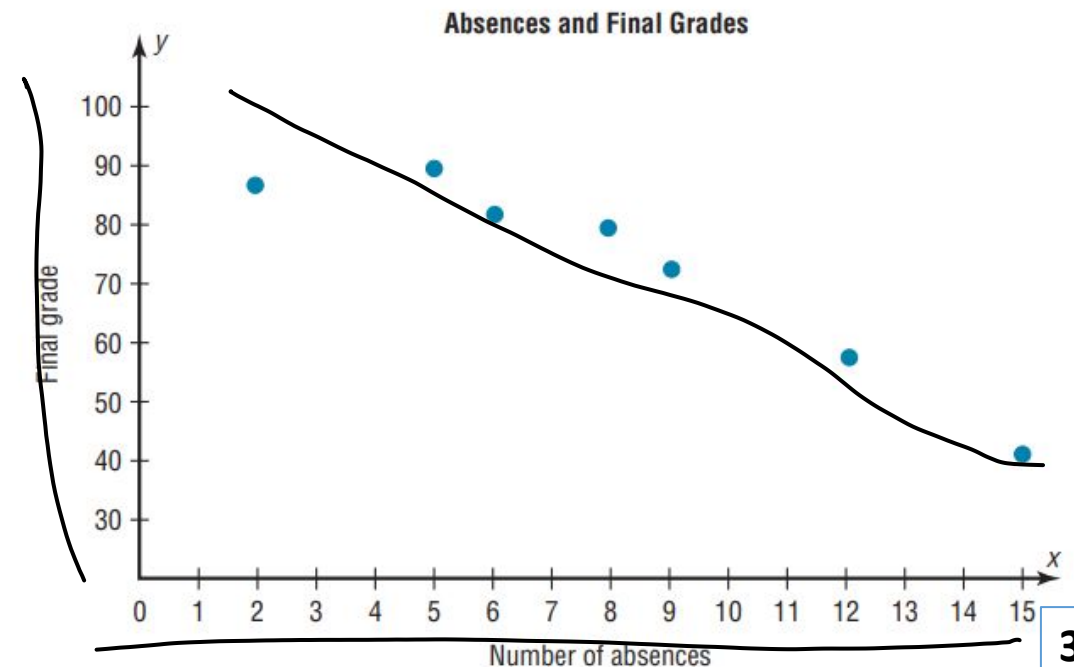
Construct a scatter plot for the data obtained in a study on the number of absences and the final grades of seven randomly selected students from a statistics class. The data are shown here.

Student	Number of absences x	Final grade y (%)
A	6	82
B	2	86
C	15	43
D	9	74
E	12	58
F	5	90
G	8	78

SOLUTION

In this example, it looks as if a negative linear relationship exists between the number of student absences and the final grade of the students.

FIGURE 10-3 Scatter Plot for Example 10-2



EXAMPLE 10-3 Age and Wealth

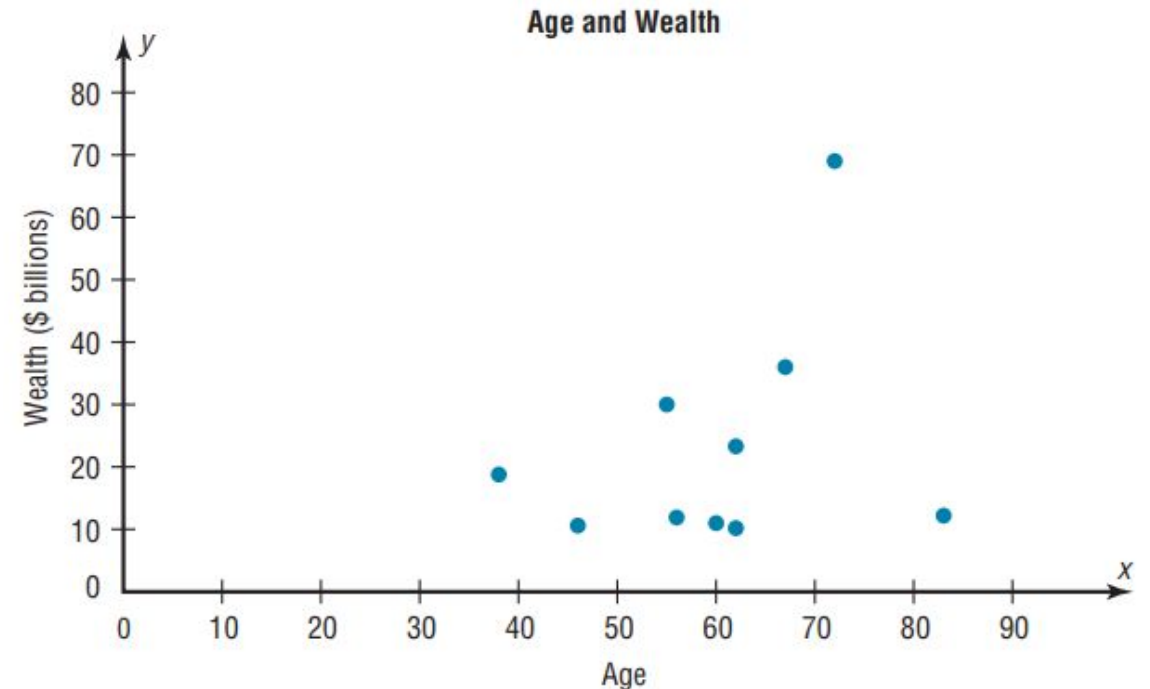
A researcher wishes to see if there is a relationship between the ages of the wealthiest people in the world and their net worth. A random sample of 10 persons was selected from the *Forbes* list of the 400 richest people for a recent year. The data are shown. Draw a scatter plot for the data.

Person	Age x	Net worth y (in billions of dollars)
A	60	11
B	72	69
C	56	11.9
D	55	30
E	83	12.2
F	67	36
G	38	18.7
H	62	10.2
I	62	23.3
J	46	10.6

SOLUTION

In this example, there is no type of a strong linear or curvilinear relationship between a person's age and his or her net worth

FIGURE 10-4 Scatter Plot for Example 10-3



Correlation

Correlation Coefficient Statisticians use a measure called the *correlation coefficient* to determine the strength of the linear relationship between two variables. There are several types of correlation coefficients.

The **population correlation coefficient** denoted by the Greek letter ρ is the correlation computed by using all possible pairs of data values (x, y) taken from a population.

The **linear correlation coefficient** computed from the sample data measures the strength and direction of a linear relationship between two quantitative variables. The symbol for the sample correlation coefficient is r .

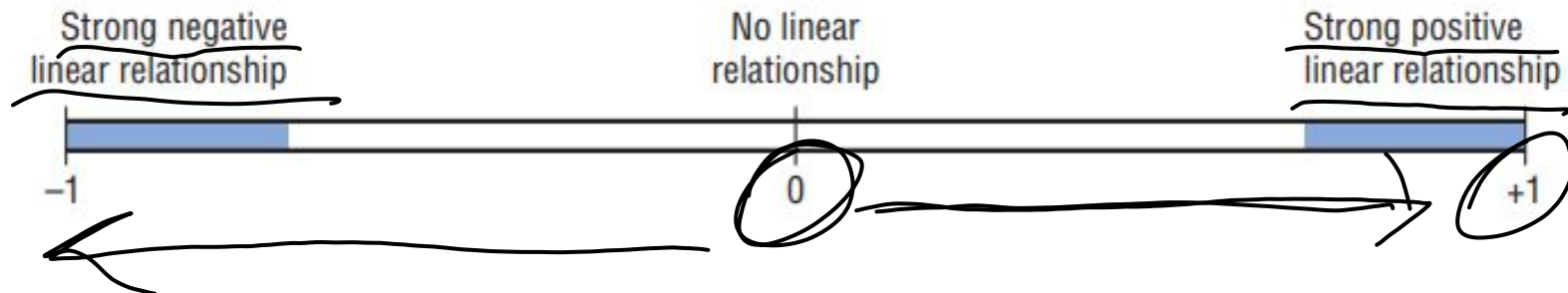
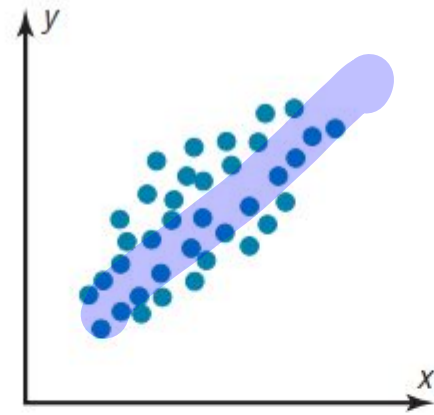
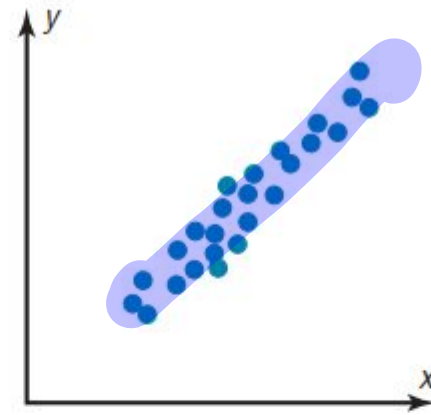


FIGURE 10-6

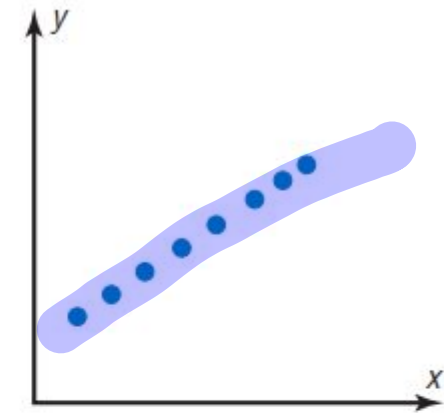
Relationship Between the
Correlation Coefficient and
the Scatter Plot



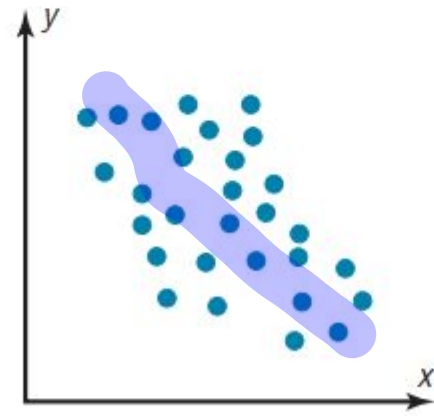
(a) $r = 0.50$



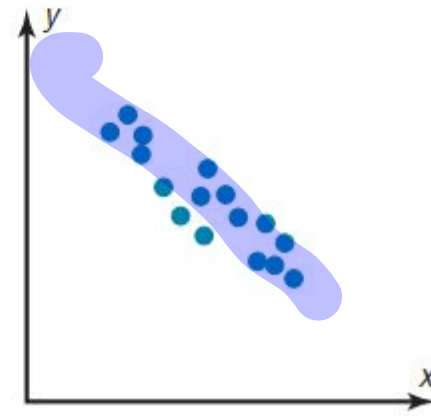
(b) $r = 0.90$



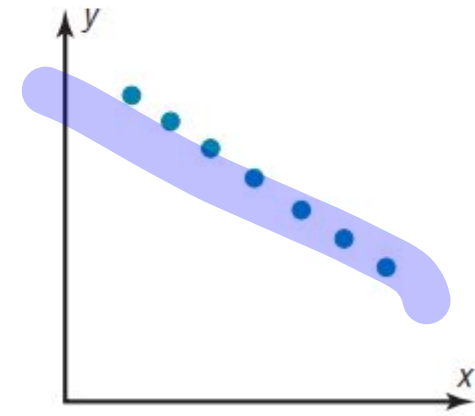
(c) $r = 1.00$



(d) $r = -0.50$



(e) $r = -0.90$



(f) $r = -1.00$

Formula for the Linear Correlation Coefficient r

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n(\sum x^2) - (\sum x)^2][n(\sum y^2) - (\sum y)^2]}}$$

where n is the number of data pairs.

EXAMPLE 10-4 Car Rental Companies

Compute the linear correlation coefficient for the data in Example 10-1.

SOLUTION

Company	Cars x (in 10,000s)	Revenue y (in billions of dollars)	xy	x^2	y^2
A	63.0	7.0	441.00	3969.00	49.00
B	29.0	3.9	113.10	841.00	15.21
C	20.8	2.1	43.68	432.64	4.41
D	19.1	2.8	53.48	364.81	7.84
E	13.4	1.4	18.76	179.56	1.96
F	8.5	1.5	12.75	72.25	2.25
	$\Sigma x = 153.8$	$\Sigma y = 18.7$	$\Sigma xy = 682.77$	$\Sigma x^2 = 5859.26$	$\Sigma y^2 = 80.67$

Substitute in the formula and solve for r :

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n(\sum x^2) - (\sum x)^2][n(\sum y^2) - (\sum y)^2]}}$$

$$= \frac{(6)(682.77) - (153.8)(18.7)}{\sqrt{[(6)(5859.26) - (153.8)^2][(6)(80.67) - (18.7)^2]}} = \underline{0.982}$$

The linear correlation coefficient suggests a strong positive linear relationship between the number of cars a rental agency has and its annual revenue. That is, the more cars a rental agency has, the more annual revenue the company will have.

EXAMPLE 10–6 Age and Wealth

Compute the value of the linear correlation coefficient for the data given in Example 10–3 for the age and wealth of the richest persons in the world.

Do yourself. Page: 558

Exercise:

13. Commercial Movie Releases The yearly data have been published showing the number of releases for each of the commercial movie studios and the gross receipts for those studios thus far. Based on these data, can it be concluded that there is a linear relationship between the number of releases and the gross receipts?

No. of releases x	361	270	306	22	35	10	8	12	21
Gross receipts y (million \$)	3844	1962	1371	1064	334	241	188	154	125

20. Water and Carbohydrates Continuing the theme of fruits and vegetables, here are the number of grams of water and the number of grams of carbohydrates for a random selection of raw foods (100 g each). Is there a linear relationship between the variables? (The information in this exercise will be used for Exercises 20 and 38 in Section 10–2.)

Water	83.93	80.76	87.66	85.20	72.85	84.61	83.81
Carbs	15.25	16.55	11.10	13.01	24.27	14.13	15.11

Source: *Time Almanac*.